CropFinder: AI-based Detection and Tracking of Crops for Precision Agriculture

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***Abstract*—Individual plant treatment, such as irrigation, fertilization, and pesticide application is pivotal in precision agriculture. Traditional methods, which detect all plants using robotic cameras and apply uniform farming practices, are not cost-effective or eco-friendly. This study introduces efficient and precise detection and tracking methods for individual lettuce plants. Our method automates the annotation of lettuce plants using the Grounded Segment Anything Model (SAM) to train the YOLOv8m detection model, achieving a mean average precision of 87%. For tracking, the study employed ByteTrack and deep observation-centric simple online and real-time tracking algorithms. The latter yielded the best results with an 87% higher order tracking accuracy due to its deep feature association methods, enabling re-identification capabilities. This algorithm also supports real-time tracking, effectively differentiating and identifying plants even during extended occlusions, such as camera retraction. Further improvements in tracking accuracy were explored using grayscale imagery and varying contrast levels in video streams, with a contrast level of 1.5 achieving an accuracy of 87.8%. Compared to existing methods, this study achieves better detection and tracking performance in agricultural settings.**

***Index Terms*—Lettuce detection, Multi-object tracking, Re-identification, YOLO, Precision agriculture**

I. Introduction

As the global population continues to surge to unprecedented levels, the demand for food production escalates, necessitating enhancements in agricultural efficiency and sustainability. Traditional agricultural practices, while foundational, often fall short in meeting contemporary demands due to their generalized and labor-intensive nature. The indiscriminate use of fertilizers, a consequence of these traditional methods, not only fails to optimize plant growth but also causes significant environmental harm, exacerbating climate change issues [1]. In this context, precision agriculture emerges as a transformative solution, aiming to revolutionize crop cultivation and management through cutting-edge technologies [2].

The advent of computer vision (CV) and robotic technologies in agriculture signifies a pivotal shift towards data-driven automatic farming practices. By harnessing these technologies, precision agriculture proposes a methodical approach to crop management, enabling the tailored application of inputs such as water, pesticides, and fertilizers at an individual plant level [3]. This precision not only enhances crop yield and quality but also mitigates the environmental impact of farming operations, marking a significant leap up in sustainable agriculture [4].

Despite the promising advantages of precision agriculture, its implementation faces significant challenges. Among the most daunting is the need for extensive and precise data on crop status across vast farmlands, a task that traditional manual methods cannot efficiently accomplish [5]. This necessitates the development of automated systems capable of collecting and processing agricultural data with minimal human intervention. Generative Adversarial Networks (GANs) and few-shot learning have been explored for their potential to generate high-quality, annotated datasets from limited initial inputs, overcoming constraints posed by small datasets and the laborious nature of manual annotation [6].

Detecting and tracking individual plants is critical for robotic systems in field farming operations, particularly for executing repetitive trajectories. While identifying plant locations allows for precise spraying of each detected plant, relying solely on detection results leaves the robot unable to track which plants have already been sprayed as it moves through the farm lanes. This necessitates the inclusion of individual plant tracking. Tracking precision is essential to avoid repeated treatment of individual plants, ensuring each plant receives the necessary care without excess, thereby minimizing waste and potential harm. Tracking often uses the Kalman filter [7] for motion prediction. The Simple Online and Real-time Tracking (SORT) algorithm [8] utilizes Kalman filtering for multi-object tracking (MOT) by associating bounding boxes in successive frames. SORT has limitations, addressed by Observation-Centric SORT (OCSORT) [9], with further improvements in deep observation-centric simple online and real-time tracking (DeepOCSORT) [10]. ByteTrack [11], another algorithm, focuses on motion-based tracking using the Kalman filter. These algorithms have been applied to various tasks, such as tracking black cattle with a 96.88% Multi-Object Tracking Accuracy (MOTA) [12]. Other research enhances YOLO for wheat head detection, achieving a mAP of 0.688 [13]. Hu et al. proposed a YOLOv5 and Kalman filter-based lettuce detection and tracking method, achieving a tracking accuracy of below 80% [14].

This study presents an enhanced approach to precision agriculture by utilizing the YOLOv8 object detection model,

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trained on automatically annotated datasets, for the accurate detection of individual plants. Additionally, it introduces DeepOCSORT, an advanced tracking algorithm, for continuous monitoring and management of plant health. The proposed approach can simultaneously detect and track lettuces, recovering the ID of a plant even if it moves out of the camera's field of view, where existing MOT methods usually fail and assign a new plant ID. The contributions of our proposed approach are as follows.

- Proposed a deep learning-based method for identifying individual plants, allowing agricultural robots to perform precision spray tasks with greater flexibility.
- Demonstrated the efficacy of automatically annotated datasets in training deep learning models for agricultural applications, reducing the reliance on labor-intensive manual annotations.
- Experimental results illustrate the effectiveness of the proposed method for lettuce detection and tracking, with potential applications for other vegetables and crops.

The paper is organized as follows: Section 2 delves into the methodology, detailing the process of data annotation, model training, and implementation. Section 3 outlines the evaluation metrics employed to assess the performance of the proposed models. Section 4 presents the results of the study, demonstrating the accuracy and efficiency of the models in agricultural settings. Finally, the conclusion summarizes the findings and proposes directions for future research in precision agriculture.

II. Material and Methodology

A. Dataset

In this study, three main datasets have been utilized: the LettuceMOT [15] dataset which contains the ground truth (GT) data annotations for tracking, LettuceMOTS [14] dataset and Lettuce Farm SLAM Dataset (LFSD) [16]. All datasets consist of RGB image sequences with a size of 810×1080. A total of 248 images from the LettuceMOTS dataset were selected to be annotated for training the detection model.

The Autodistill library[1], which enables the users to transfer the knowledge of complex foundational modules to much simpler models for specific downstream tasks has been used in this work to fully automate the annotation process for the training and, hence achieving the goal with minimal manual intervention. Foundational models like Meta's segment anything model (SAM) trained with 11 million images and 1.1 billion masks demonstrate zero-shot abilities. SAM's prompts include points, bounding boxes, and texts, producing masks with confidence values. Compared to existing methods, SAM excels in medical image analysis [17]. Studies explore SAM's application in counting objects and suggest combining reference object selection for better results [18]. To enhance text-prompted segmentation, Grounding DINO is proposed [19]. SAM adapters are recommended for medical image segmentation to address lackluster results in specific tasks [20], contributing to zero-shot learning in downstream tasks with fewer or no images.

In our study, the text prompt, 'Lettuce plants' is given to the Grounding DINO which creates the bounding boxes and gives the output to SAM to generate the segmentation masks along with the annotations. The automatically acquired detection bounding boxes along with the segmentation annotations for the 3 growth stages can be visualized as shown in Fig. 1 where the text prompt was 'Lettuce plants'. As it shows, the automatic annotations give the same results as if it was done manually.

B. YOLOv8 Based Plant Detection

YOLO is a popular object detector that trades some accuracy for speed [21]. Various YOLO versions are widely used in applications like fruit [22] and face detection [23]. YOLOv8, an improvement over YOLOv5, is used for flying object detection [**?**], showcasing an interesting approach for training.

The YOLOv8m excels in real-time object detection. It segments an image into a grid, with each cell responsible for predicting object boundaries and class likelihoods. Through successive iterations, YOLOv8m has enhanced its speed, precision, and sturdiness, adeptly managing the detection of multiple objects with its anchor box system and intersection over union (IoU) technique, ensuring the selection of the most accurate bounding box for object identification. This versatility makes YOLOv8m suitable for a variety of real-time detection applications. Therefore, the YOLOv8m including its Spatial Pyramid Pooling Fusion (SPPF), backbone and detect architectures were used to train the lettuce detection model with the annotated dataset.

C. Tracking Integration

This subsection discusses the implementation of the ByteTrack and DeepOCSORT algorithms for tracking. It was checked for different contrast levels of the image frames, grayscale, contrast level with 0.5 and 1.5. All the videos were of a 24 real-time frame rate. The trained detection model was utilized to track the plants individually. Each image frame is given as an input to the trained YOLOv8 detection model and the detected bounding boxes are then given to the tracking algorithms. Although YOLOv8 itself offers in-built tracking, it does not consider features of the objects which is why DeepOCSORT tracking algorithm is employed in this study.

1) ByteTrack based individual plant tracking: This algorithm is a motion-based tracking algorithm that does not contain reID capabilities inherited from its core implementation unless a reID module is separately integrated. Employing the Kalman Filter as its foundational piece to locate the position of an object in frame i+1 after detecting and identifying it in frame i as well, ByteTracks innovation is that it does not disregard the tracklets that are not matched without checking them against detections with low confidence scores. The track_thresh, track_buffer and match_thresh which were used to filter out tracklets, to count the number of frames a track is lost and to set Intersection over Union (IoU) threshold as

[1]https://github.com/autodistill

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the matching threshold respectively. These parameters were fine-tuned and the combination of 0.7, 30 and 0.9 for the above parameters was identified as the optimal solution after evaluating results for each finetuning combination.

2) DeepOCSORT based individual plant tracking: OCSORT is an improvement over SORT in that it computes a virtual trajectory for occlusion periods (especially when nonlinear motion is also occurring simultaneously) using detector measurements rather than depending only on the state estimation, thereby minimizing the escalation of errors that result in missing trajectories during periods of occlusion [12]. Further to this, OCSORT also incorporates another correction, the direction consistency of tracks named Observation-Centric Momentum. Faults in detections and camera blur can introduce errors in tracking which can be identified as limitations of this algorithm. DeepOCSORT introduces corrections for camera motion blur to increase detection accuracy and to improve the discriminative nature of the appearance [13]. In the DeepOCSORT tracking algorithm, tracking is made more accurate than OCSORT by introducing 3 corrections, Camera Motion Compensation (CMC), Dynamic Appearance (DA), and Adaptive Weighting (AW).

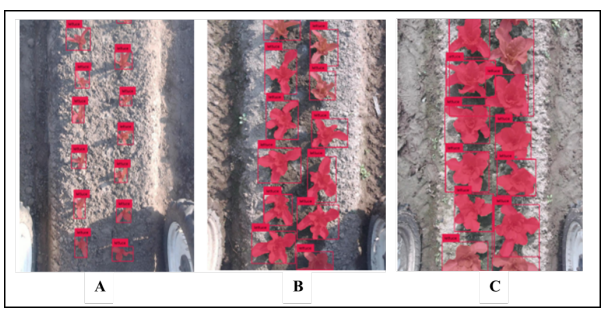

Fig. 1. Grounded SAM bounding box and segmentation annotations (A) Growth Stage 1 (B) Growth Stage 2 (C) Growth Stage 3.

III. Experimental setup

The training of the detection model along with the implementation of the tracking module and inferencing of the videos including testing and evaluation was done using Google Colaborotary and the Tesla T4 GPU offered by it.

For the detection model training, the input image size was 810×1080 and the batch size was set to 32 while the optimizer and the learning rate were set to SGD and 0.01 respectively when training the model for 50 epochs. Out of the 248 images, 198 were used for training while 50 were used for validation. These parameters resulted in the most optimal solutions they were fine-tuned in several combinations.

For the implementation of the DeepOCSORT, the BoxMOT [24] implementation of SOTA tracking algorithms has been employed. The following parameters were modified during the process of finding the best implementation, model_weights, det_thresh, max_age, min_hits, iou_threshold, asso_func, w_association_emb, embedding_off, cmc_off, aw_off, new_kf_off. CMC is a correction to make up for any errors due to camera motion that changes the position of the center and coordinates of the detection boxes using two matrices to incorporate the corrections. AW on the other hand, is a parameter defined to improve appearance matching between detection and tracking boxes. This helps to discriminate between objects of quite similar appearances. For the CMC that is defined above, the Enhanced Correlation Coefficient (ECC) affine correction method was selected to improve results for the re-identification of plants that have left the frame and reappeared later on.

IV. Performance Evaluation

The evaluation of this research has been done in two parts, where the detection model was first evaluated using the Mean Average Precision (mAP), recall, precision and F1 score.

The model that was combined with the tracking method was evaluated using the higher order tracking accuracy (HOTA) metrics [25]. For this, TrackEval [26] MOTChallenge Kit has been utilized. This research specifically utilizes TrackEval's Higher Order Tracking Accuracy (HOTA) metrics and Identity metrics. HOTA assesses the balance between object detection and association efficiency, while Identity metrics gauge identity maintenance across frames in videos or image sequences. HOTA is a versatile metric offering a unified measure for ranking tracking systems, breaking down into submetrics for detailed analysis of tracking aspects. It unifies three IoU scoresdetection, association, and localizationpreviously not fully considered in evaluation metrics.

Computed against various localization accuracy thresholds (α), HOTA provides a comprehensive assessment of tracking performance. The concept of True Positives (TP), False Positives (FP) and False Negatives (FN) is employed in tracking as well. A TP can be defined as a pair that includes both a ground truth detection and a predicted detection, where the localization similarity (S) meets or exceeds the specified threshold, localization threshold value α. for each TP, True Positive Associations (TPA) and False Positive Associations (FPA) and False Negative Associations (FNA) have been defined. A set of TPAs is defined as for a c (an instance of a TP), a set that includes all TPs that have matching ground truth detections and predicted detections like c. Association Accuracy is referred to as AssA, and is the mean alignment between paired trajectories, computed by considering all the detections and averaging across them.

Detection Accuracy is represented as DetA. Localization accuracy (LocA) thresholds (α) represent different LocA thresholds from 0 to 1. The optimal balance point, where both detection and tracking evaluations peak before dropping off, occurs at (α). Equation (1) represents the HOTA_α, HOTA value for each localization threshold from 0 to 1.

The final HOTA value as shown in (1) is calculated as the geometric mean between the detection accuracy and the association accuracy. This computation is performed across various localization thresholds which is how all three values are taken into consideration in this HOTA metrics.

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$$\text{HOTA}_\alpha = \sqrt{\text{DetA}_\alpha \cdot \text{AssA}_\alpha} \tag{1}$$

$$\therefore \text{HOTA} = \sqrt{\text{DetA} \cdot \text{AssA}} \tag{2}$$

The TrackEval kit offers the Identity Recall (IDR), Identity Precision (IDP) and Identity F1 score (IDF1) metrics computations. IDTP, IDFP and IDFN refer to True Positive ID, False Positive ID and False Negative ID respectively [27].

V. Results

The results are mainly categorized into two subsections, results obtained for object detection and object tracking.

A. Object Detection Training Results

The YOLOv8m model was selected as the model that produced the best results for the detection model. The model comparison for the evaluation metrics, mAP@0.5, precision, recall, F1 Score are shown in Table 1.

The YOLOv8m model outperforms its nano counterpart (YOLOv8n), justifying its suitability for tracking implementation despite being nearly 10 times larger. Prioritizing accuracy over model size is crucial for detecting and tracking lettuce plants accurately across frames. YOLOv8m's confidence threshold of 0.7 strikes a balance, retaining true positives without an excessive loss of false positives. YOLOv8n, with a higher threshold (0.95), loses some true positives. The difference in F1 scores and recall values between the two models in Table I supports this observation. Expanding the YOLOv8n model's capabilities for tracking would require dataset augmentation, potentially incorporating diverse data from another dataset.

TABLE I
Performance Metrics for Different Models

Model	mAP@0.5	Precision	Recall	F1 Score
YOLOv8n	0.79	0.90	0.77	0.61
YOLOv8m	0.87	0.91	0.83	0.87

B. Object Tracking Results

Tracking results for both ByteTrack and DeepOCSORT algorithms are discussed. TrackEval's graphs display evaluation metric scores on the y-axis, with localization accuracy thresholds (α) on the x-axis, representing different LocA thresholds from 0 to 1. The optimal balance point, where both detection and tracking evaluations peak before dropping off, occurs at α. Overall metric measurements (HOTA, DetA, AssA, DetRe, DetPr, AssRe, AssPr) are obtained by calculating the area under each curve, in the graph. The lower AssA suggests challenges in re-identification when plants leave and re-enter the frame, impacting the assignment of consistent object IDs.

Integrated with the observation-centric approach of the SORT tracking algorithm and appearance features, the BoxMOT DeepOCSORT tracking module [24] was aimed to address re-identification issues. The evaluation covered the original, grayscale, contrast set to 0.5, and contrast set to 1.5 videos. These contrast levels have been selected to have the results to be compared easily with the similar study [28] that has been done.

In Fig. 2, HOTA evaluation metrics are plotted against LocA threshold values (α) for all four types of videos. Comparing the HOTA graph for the original video with others, the video with contrast set to 1.5 shows slight variations. DeepOCSORT on grayscale exhibited the lowest performance, indicating color's role in detection and tracking. Contrast modification (0.5 and 1.5) impacted tracker performance, discussed further in the next subsection. A basic counter was implemented to ensure accurate plant count (disregarding re-identification). The counter was displayed at the top of the frame, showcasing results with visible object IDs. As every detected plant is assigned an ID before the confidence score is applied, some falsely detected plants are assigned IDs and these do not appear in the frames once filtered through the confidence thresholds, hence some IDs appear to be skipped. Therefore, highest value ID isn't a reliable final count; thus, the displayed count accurately reflects the correct plant count.

Fig. 3 illustrates the visualizations of the DeepOCSORT tracker on straight1 (Fig. 3-A), straight2 (Fig. 3-B) as well as on the BF2 image sequence. Fig. 3-C illustrates how the tracker has performed on the BF2 sequence when the detection

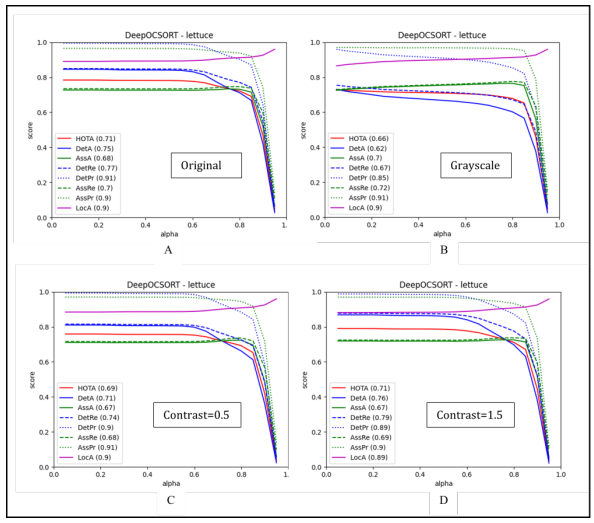

Fig. 2. HOTA Evaluation Graphs (A) Graph for Original Video (B) Graph for Grayscale video (C) Graph for Contrast set to 0.5 video (D) Graph for Contrast set to 1.5 video.

threshold is set to 0.7 while Fig. 3-D shows how the detections have gotten better when the threshold was decreased to 0.3 indicating how different confidence levels impact the detection of the plants. Having a higher confidence level is disadvantageous when plants detected and assigned IDs in a previous frame are detetcted with a detection accuracy lower than the threshold in a particular frame and are disregarded at that time but the same plants are detected in a future frame and are assigned different IDs. The difference in the lettuce count is due to this dicrepancy.

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TABLE II
EVALUATION METRICS COMPARISON OF BYTETRACK AND DEEPOCSORT

Dataset	Method	HOTA (%)	DetA (%)	AssA (%)	AssRe (%)	AssPr (%)	IDF1 (%)	IDR (%)	IDP (%)
Straight1	ByteTrack	86.29	86.27	86.45	88.01	94.84	94.46	93.05	95.92
	DeepOCSORT	87.19	86.27	88.22	89.40	93.87	97.05	95.05	99.14
B&F1	ByteTrack	65.55	86.11	49.95	50.44	94.33	58.39	57.62	59.18
	DeepOCSORT	66.59	85.19	52.12	53.76	89.69	62.62	60.59	64.79
Straight2	ByteTrack	83.51	84.60	82.53	84.00	93.60	91.76	89.50	94.15
	DeepOCSORT	86.38	85.00	87.88	88.83	94.00	96.04	93.61	98.60
B&F2	ByteTrack	35.57	43.87	28.93	29.35	88.67	38.51	28.79	58.16
	DeepOCSORT	41.17	60.99	33.60	35.99	80.51	47.42	38.52	61.66
Combined	ByteTrack	69.54	73.56	65.92	67.05	93.54	71.61	65.44	79.07
	DeepOCSORT	71.44	75.09	68.13	69.62	90.33	75.99	70.44	82.50

TABLE III
COMPARISON FOR DEEPOCSORT APPLIED TO GRAYSCALE AND CONTRAST-ENHANCED IMAGE SEQUENCES

Dataset	Image Type	HOTA (%)	DetA (%)	AssA (%)	AssRe (%)	AssPr (%)	IDF1 (%)	IDR (%)	IDP (%)
Straight1	Grayscale	85.61	84.56	86.80	87.99	94.06	96.03	93.17	99.06
	Contrast factor = 0.5	84.91	83.81	86.12	87.28	93.59	95.88	92.92	99.03
	Contrast factor = 1.5	87.78	85.96	88.70	89.93	93.74	97.51	95.74	99.34
	Original	87.19	86.27	88.22	89.40	93.87	97.05	95.05	99.14
B&F1	Grayscale	65.39	86.03	49.75	51.06	91.38	59.84	57.64	62.21
	Contrast factor = 0.5	65.96	82.80	52.63	54.24	89.76	63.38	60.81	66.18
	Contrast factor = 1.5	66.07	86.40	50.57	51.16	92.97	59.92	58.37	61.55
	Original	66.59	85.19	52.12	53.76	89.69	62.62	60.59	64.79
Straight2	Grayscale	85.49	84.34	86.80	87.59	93.94	95.32	92.54	98.27
	Contrast factor = 0.5	83.30	81.86	84.87	85.81	93.73	94.24	90.50	98.29
	Contrast factor = 1.5	86.95	85.76	88.23	89.18	93.77	96.00	94.17	97.90
	Original	86.38	85.00	87.88	88.83	94.00	96.04	93.61	98.60
B&F2	Grayscale	15.10	15.13	15.69	16.77	65.25	16.54	11.76	27.85
	Contrast factor = 0.5	35.52	44.75	28.59	30.03	81.97	41.54	32.53	57.48
	Contrast factor = 1.5	41.04	53.10	31.96	34.39	75.79	46.89	40.33	56.01
	Original	41.17	60.99	33.60	35.99	80.51	47.42	38.52	61.66
Combined	Gray scale	65.87	61.79	70.41	71.57	91.45	68.58	61.27	77.88
	Contrast factor = 0.5	68.78	71.32	66.57	67.86	90.65	74.10	67.47	82.18
	Contrast factor = 1.5	71.38	75.89	67.41	68.71	90.39	74.93	70.76	79.62
	Original	71.44	75.09	68.13	69.62	90.33	75.99	70.44	82.50

Fig. 4 illustrates how DeepOCSORT has been able to re-identify some of the plants that had left the frame and reappeared later on. Fig. 4-A shows the IDs given to each plant in forward motion while Fig. 4-B shows the IDs given to each plant in backward motion.

DeepOCSORT generally outperformed ByteTrack as shown in Table II, especially in straight image sequences, although the difference is marginal. DeepOCSORT's consideration of feature appearances contribute to its slightly superior results. BF2 exhibited the worst performance, due to its higher illumination level compared to other sequences. The combination of YOLOv8 with DeepOCSORT in straight sequences surpassed a previous lettuce crop detection study [28], achieving an 87% HOTA score compared to the previous study's 78%. However, the study's introduction of a geometrical feature yielded better results for BF image sequences.

Table III provides a comparison between original, grayscale, decreased (set to 0.5) or increased (set to 1.5) contrast image sequences. In straight motion sequences, higher contrast levels yielded improved results, whereas in BF sequences, the original images slightly outperformed the contrast-enhanced ones,

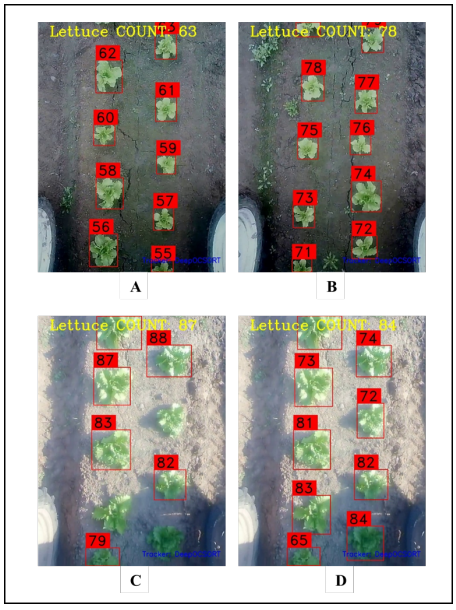

Fig. 3. DeepOCSORT Visualization for Image Sequences/Videos (A) straight1 (B) straight2 (C) BF2 with conf=0.7 (D) BF2 with conf=0.3.

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although the difference is minimal.

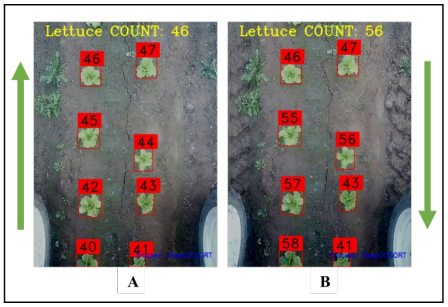

Fig. 4. DeepOCSORT for BF1 (A) forward motion (B) backward motion.

VI. Conclusion

To enable precise individual plant treatments, this study presents a YOLOv8m-based detection approach for identifying lettuce plants from video streams or image sequences. Subsequently, ByteTrack and DeepOCSORT algorithms were applied for the individual tracking of lettuce plants. The YOLOv8m model, trained for this purpose, exhibited a mAP of 87% at the 0.5 IoU threshold, proving to be both effective and accurate for detecting lettuce. Notably, the DeepOCSORT algorithm outperformed previous research on the same dataset, achieving a Higher Order Tracking Accuracy (HOTA) of 87%.

This study highlights the potential for further development, particularly through the exploration of tracking performance under various lighting conditions. It was observed that both tracking algorithms encountered decreased performance in image sequences with high contrast compared to other sequences. Moreover, the methodology applied here demonstrates the potential for broader agricultural applications, extending beyond lettuce to crops such as corn, wheat, and sugar beet.

References

[1] M. Roser, "Employment in agriculture," *Our World in Data*, 2023, https://ourworldindata.org/employment-in-agriculture.

[2] I. Cisternas, I. Velásquez, A. Caro, and A. Rodríguez, "Systematic literature review of implementations of precision agriculture," *Computers and Electronics in Agriculture*, vol. 176, p. 105626, 2020.

[3] J. Kaur and W. Singh, "Tools, techniques, datasets and application areas for object detection in an image: a review," vol. 81, no. 27, pp. 38 297–38 351. [Online]. Available: https://doi.org/10.1007/s11042-022-13153-y

[4] Y. Qiao, J. Valente, Y. Jiang, Z. Zhang, D. He, and D. Su, *AI, Sensors and Robotics in Plant Phenotyping and Precision Agriculture, Volume II*. Frontiers Media SA, 2023.

[5] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surv.*, vol. 53, no. 3, jun 2020. [Online]. Available: https://doi.org/10.1145/3386252

[6] C. Zhou, Z. Zhang, S. Zhou, J. Xing, Q. Wu, and J. Song, "Grape leaf spot identification under limited samples by fine grained-gan," *IEEE Access*, vol. 9, pp. 100 480–100 489, 2021.

[7] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 03 1960. [Online]. Available: https://doi.org/10.1115/1.3662552

[8] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *2016 IEEE International Conference on Image Processing (ICIP)*. IEEE, Sep. 2016. [Online]. Available: http://dx.doi.org/10.1109/ICIP.2016.7533003

[9] J. Cao, J. Pang, X. Weng, R. Khirodkar, and K. Kitani, "Observation-centric sort: Rethinking sort for robust multi-object tracking," 2023.

[10] G. Maggiolino, A. Ahmad, J. Cao, and K. Kitani, "Deep oc-sort: Multi-pedestrian tracking by adaptive re-identification," in *2023 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2023, pp. 3025–3029.

[11] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "Bytetrack: Multi-object tracking by associating every detection box," in *European conference on computer vision*. Springer, 2022, pp. 1–21.

[12] S. Myat Noe, T. T. Zin, P. Tin, and I. Kobayashi, "Comparing state-of-the-art deep learning algorithms for the automated detection and tracking of black cattle," *Sensors*, vol. 23, no. 1, 2023. [Online]. Available: https://www.mdpi.com/1424-8220/23/1/532

[13] Y. Zhang, M. Li, X. Ma, X. Wu, and Y. Wang, "High-precision wheat head detection model based on one-stage network and gan model," *Frontiers in Plant Science*, vol. 13, 2022. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fpls.2022.787852

[14] NanH. LettuceMOTS. Original-date: 2023-06-02T13:08:09Z. [Online]. Available: https://github.com/NanH5837/LettuceMOTS

[15] N. Hu, S. Wang, X. Wang, Y. Cai, D. Su, P. Nyamsuren, Y. Qiao, Y. Jiang, B. Hai, and H. Wei, "Lettucemot: A dataset of lettuce detection and tracking with re-identification of re-occurred plants for agricultural robots," *Frontiers in Plant Science*, vol. 13, 2022. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fpls.2022.1047356

[16] wangshuo9707. LFSD-dataset. Original-date: 2023-02-27T12:47:03Z. [Online]. Available: https://github.com/wangshuo9707/LFSD-Dataset

[17] C. Mattjie, L. V. de Moura, R. C. Ravazio, L. S. Kupssinsk, O. Parraga, M. M. Delucis, and R. C. Barros, "Zero-shot performance of the segment anything model (sam) in 2d medical imaging: A comprehensive evaluation and practical guidelines," 2023.

[18] Z. Shi, Y. Sun, and M. Zhang, "Training-free object counting with prompts," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 323–331.

[19] L. P. Osco, Q. Wu, E. L. de Lemos, W. N. Gonalves, A. P. M. Ramos, J. Li, and J. Marcato, "The segment anything model (sam) for remote sensing applications: From zero to one shot," *International Journal of Applied Earth Observation and Geoinformation*, vol. 124, p. 103540, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1569843223003643

[20] J. Wu, Y. Zhang, R. Fu, H. Fang, Y. Liu, Z. Wang, Y. Xu, and Y. Jin, "Medical sam adapter: Adapting segment anything model for medical image segmentation," 2023.

[21] P. Jiang, D. Ergu, F. Liu, Y. Cai, and B. Ma, "A review of yolo algorithm developments," *Procedia Computer Science*, vol. 199, pp. 1066–1073, 2022, the 8th International Conference on Information Technology and Quantitative Management (ITQM 2020 & 2021): Developing Global Digital Economy after COVID-19. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050922001363

[22] C. Liang, J. Xiong, Z. Zheng, Z. Zhong, Z. Li, S. Chen, and Z. Yang, "A visual detection method for nighttime litchi fruits and fruiting stems," *Computers and Electronics in Agriculture*, vol. 169, p. 105192, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169919313274

[23] Y. V. Kale, A. U. Shetty, Y. A. Patil, R. A. Patil, and D. V. Medhane, "Object detection andface recognition using yolo andinception model," in *Advanced Network Technologies and Intelligent Computing*, I. Woungang, S. K. Dhurandher, K. K. Pattanaik, A. Verma, and P. Verma, Eds. Cham: Springer International Publishing, 2022, pp. 274–287.

[24] M. Brostrm. (2023, Feb.) Real-time tracking and segmentation using yolov8. [Online]. Available: https://doi.org/10.5281/zenodo.7629840

[25] J. Luiten, A. Osep, P. Dendorfer, P. Torr, A. Geiger, L. Leal-Taix, and B. Leibe, "HOTA: A higher order metric for evaluating multi-object tracking," vol. 129, no. 2, pp. 548–578. [Online]. Available: https://doi.org/10.1007/s11263-020-01375-2

[26] J. Luiten. JonathonLuiten/TrackEval: HOTA (and other) evaluation metrics for multi-object tracking (MOT). [Online]. Available: https://github.com/JonathonLuiten/TrackEval

[27] E. Ristani, F. Solera, R. Zou, R. Cucchiara, and C. Tomasi, "Performance measures and a data set for multi-target, multi-camera tracking," in *European conference on computer vision*. Springer, 2016, pp. 17–35.

[28] N. Hu, D. Su, S. Wang, P. Nyamsuren, Y. Qiao, Y. Jiang, and Y. Cai, "Lettucetrack: Detection and tracking of lettuce for robotic precision spray in agriculture," *Frontiers in Plant Science*, vol. 13, 2022. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fpls.2022.1003243

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